

The Dot Product

①

The Dot Product of 2 vectors is a "sophisticated" means to combine vectors.

The Dot Product is defined as

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 \quad (\text{component})$$

(OR)

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos(\theta), \quad (\text{geometric})$$

where θ is the angle between $\vec{u}$ and $\vec{v}$ and $0^\circ \leq \theta \leq 180^\circ$.

NOTES: ① If either $\vec{u}$ or $\vec{v}$ is a zero vector then $\vec{u} \cdot \vec{v} = 0$.

② If $\vec{u}$ and $\vec{v}$ are $\perp$ then $\vec{u} \cdot \vec{v} = 0$
since $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \overset{0}{\cos(90^\circ)}$
 $= 0$

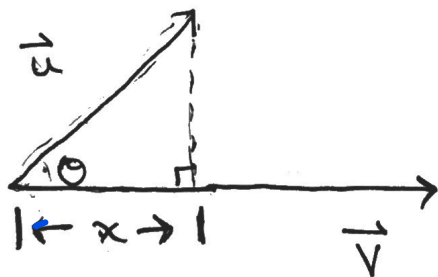
③ The dot product is a scalar not a vector.

$$\textcircled{4} \quad \vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$$

$$\textcircled{5} \quad \vec{u} \cdot \vec{u} = |\vec{u}|^2 \quad (\cos(0^\circ) = 1)$$

$$\textcircled{6} \quad \vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} \quad \left| \begin{array}{l} \text{distributive} \\ \text{law.} \end{array} \right.$$

$\textcircled{7}$ Geometrically, the dot product is the product of the length of one vector with the component of the other along the 1st vector.



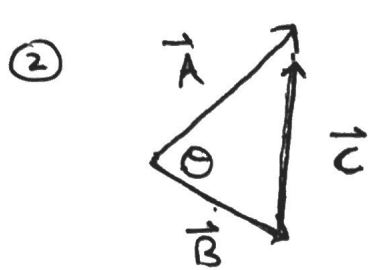
$$x = |\vec{u}| \cdot \cos(\theta)$$

$$\vec{u} \cdot \vec{v} = |\vec{u}| \cos(\theta) \cdot |\vec{v}|$$

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cdot \cos(\theta)$$

Proof of geometric definition

$$\begin{aligned} \textcircled{1} \quad \vec{A} \cdot \vec{A} &= |\vec{A}| |\vec{A}| \overset{=1}{\cos 0^\circ} \\ &= |\vec{A}|^2 \end{aligned}$$



$$\text{Let } \vec{C} = \vec{A} - \vec{B}$$

By Cosine Law:

$$|\vec{C}|^2 = |\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}||\vec{B}|\cos(\theta) \quad \textcircled{1}$$

$$|\vec{C}|^2 = \vec{C} \cdot \vec{C}$$

$$= (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B})$$

$$= \vec{A} \cdot \vec{A} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B}$$

$$= |\vec{A}|^2 + |\vec{B}|^2 - 2 \underbrace{\vec{A} \cdot \vec{B}} \quad \textcircled{2}$$

$$= |\vec{A}||\vec{B}|\cos(\theta)$$

from cosine law

$$\therefore \vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos(\theta)$$

Proof of component definition

Given $\vec{u} = [u_1, u_2]$ & $\vec{v} = [v_1, v_2]$

$$[u_1, u_2] = u_1 \hat{i} + u_2 \hat{j}$$

$$[v_1, v_2] = v_1 \hat{i} + v_2 \hat{j}$$

$$\vec{u} \cdot \vec{v} = (u_1 \hat{i} + u_2 \hat{j}) \cdot (v_1 \hat{i} + v_2 \hat{j})$$

$$\vec{u} \cdot \vec{v} = u_1 \hat{i} \cdot v_1 \hat{i} + u_1 \hat{i} \cdot v_2 \hat{j} + u_2 \hat{j} \cdot v_1 \hat{i} + u_2 \hat{j} \cdot v_2 \hat{j}$$

$$\vec{u} \cdot \vec{v} = u_1 v_1 |\hat{i}|^2 + u_1 v_2 \hat{i} \cdot \hat{j} + u_2 v_1 \hat{j} \cdot \hat{i} + u_2 v_2 |\hat{j}|^2$$

$$\vec{u} \cdot \vec{v} = u_1 v_1 (1)^2 + 0 + 0 + u_2 v_2 (1)^2$$

↑
magnitude
= 1

⊥ ⇒ cos(90°)
= 0

↑
magnitude
= 1

$\therefore \vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$

Ex 1: Find the dot product of $\vec{u}$ and $\vec{v}$ given

a) $|\vec{u}| = 6$, $|\vec{v}| = 7$ and $\theta = 45^\circ$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= |\vec{u}| \cdot |\vec{v}| \cos(\theta) \\ &= (6)(7) \cos(45^\circ) \\ &= \frac{42}{\sqrt{2}} \\ &\approx 29.7\end{aligned}$$

b) $|\vec{u}| = 15$, $|\vec{v}| = 3$ and $\theta = \frac{\pi}{6}$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= (15)(3) \cos\left(\frac{\pi}{6}\right) \\ &= 45 \frac{\sqrt{3}}{2} \\ &\approx 38.97\end{aligned}$$

c) $\vec{u} = [2, -4]$, $\vec{v} = [-7, -5]$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= (2)(-7) + (-4)(-5) \\ &= -14 + 20 \\ &= 6\end{aligned}$$

(6)

Ex 2: Determine the angle between each vector

if $\vec{u} = [-3, 6]$ and $\vec{v} = [4, 2]$

$$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cos(\theta)$$

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}$$

$$\cos(\theta) = \frac{-12 + 12}{\sqrt{45} \cdot \sqrt{20}}$$

$$\cos(\theta) = 0$$

$$\theta = \frac{\pi}{2} \quad (\text{vectors are } \perp \text{ to each other})$$

Ex 3: Simplify $(2\vec{u} - 3\vec{v}) \cdot (\vec{u} + 4\vec{v})$

$$(2\vec{u} - 3\vec{v}) \cdot (\vec{u} + 4\vec{v}) = 2\vec{u} \cdot \vec{u} + 8\vec{u} \cdot \vec{v} - 3\vec{u} \cdot \vec{v} - 12\vec{v} \cdot \vec{v}$$

$$= 2|\vec{u}|^2 + 5\vec{u} \cdot \vec{v} - 12|\vec{v}|^2$$